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Experimental Evaluation of Deep Learning Assisted Autism Spectrum Disorder Prediction Scheme using Convolutional Classification Principle

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Abstract— Autism Spectrum Disorder (ASD) is characterized by challenges in social communication and restricted or repetitive behaviors. With rising ASD incidence, early diagnosis and intervention become increasingly crucial for improving developmental trajectories and quality of life. Traditional diagnostic methods are often subjective and limited. This study evaluates a deep learning-based approach for ASD prediction using Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). Leveraging the Autism Brain Imaging Data Exchange (ABIDE) dataset, which includes neuroimaging, genetic, and behavioral data, the model integrates spatial and temporal features to enhance prediction accuracy. The combined feature set yielded the highest accuracy of 93.7%, demonstrating the efficacy of integrating multiple data types. The proposed Convolutional Recurrent Hybrid (CRH) model achieved a superior accuracy of 98%, outperforming traditional models. This research highlights the potential of deep learning techniques in improving ASD diagnostics.

***Keywords*— Autism Spectrum Disorder, deep learning, Convolutional Neural Networks, Recurrent Neural Networks, Neuroimaging, Genetic Data, Behavioral Data, Early Diagnosis, Prediction Accuracy.**

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder identified by difficulties in social communication, restrictive or repetitive behaviors and strengths and differences across a wide spectrum [1] [2]. As the incidence of ASD continues to increase, it becomes more and more important in anticipation for early intervention. By diagnosing ASD early and intervening significantly the developmental trajectory of people with autism can be positively affected, improving their quality of life as well as allowing better long-term outcomes. Hence, one of the most important topics in research and clinical practice is to find accurate prediction methods. Machine-learning and artificial intelligence have significantly transformed the prediction of ASD in recent years. Behaviors are also imprecise markers, and human behavior is shaped by multiple influences, leading to the possibility of subjective judgments in evaluation especially when traditional diagnostic methods that have so far been attributed unlimited trustworthiness are used [3] [4]. This is where machine learning algorithms can have an edge as they are able to process large, intricate data sets and look for the faintest of signals that might go unnoticed by human minds. They found that using those inputs, along with genetic information and additional data on other biological markers derived from neuroimaging or behavioral assessments in some predictions led to a nearly perfect prediction of the probability of an ASD.

The integration of multiple data types is an important component in ASD prediction. Given that so many genes are correlated with ASD from genetic studies, there is quite a bit of evidence to suggest a strong hereditary foundation for the syndrome [5] [6]. Neuroimaging techniques reveal atypical brain structures/functioning of patients with ASD. Finally, global behavioral assessments such as standardized tests and observational data provide important information regarding the early signs of autism at a developmental level. People with Autism have difficulty in understanding verbal and non-verbal communication which might be difficult to engage or form inhibition based relationships. For example, communication problems may manifest as the child having delayed speech development or a lack of verbal language [7] [8]. So eye contact is a big barrier for many people with ASD, also understanding feelings in themselves and others. Moreover, repetitive movements, but also preferences for certain routines as well some topics or objects, are quite often pronounced. Similarly, sensory sensitivities such as moderate or severe reaction to stimuli like lights, sound or texture are common. Figure 1 shows the symptoms of ASD.

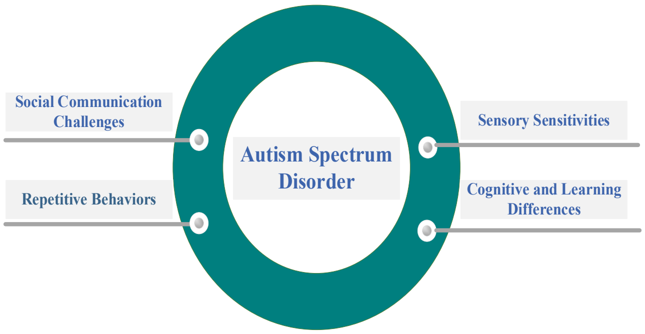

Figure 1. Symptoms of Autism Spectrum Disorder

Although there is no cure for ASD, early intervention and a combination of therapies can really make a difference in the quality of life they live. Treatment involves tailor-made programs that correspond to the needs of individuals and it typically includes behavioral interventions, speech/language therapy, occupational therapy and social skills training [9] [10]. Behavioral therapies such as Applied Behavior Analysis (ABA) and Pivotal Response Treatment (PRT), are generally considered two major methods of behavior-based therapy that aim to enhance targeted behaviors, by working on specific developmental areas through reinforcement techniques.

Convolutional neural networks are another type of deep learning model that has been structured for interpreting images, such as identifying patterns in these images. In the application of ASD prediction, CNN is used to learn geometrical feature representations from neuroimaging modalities such as magnetic resonance imaging (MRI) and functional MRI (fMRI). These scans generate high-resolution images of brain structure and function to show distinct patterns that are related with Autism Spectrum Disorder. The CNN takes these images and it first processes them through a series of convolutional layers, which detect what the data looks like such as edges, textures or shapes in an image and then pooling layer that reduces dimensionality but keeps the important information. Traditional diagnostic methods are subjective and limited in processing large, complex datasets. This study aims to evaluate the effectiveness of deep learning models, specifically CNNs and RNNs, in predicting ASD by integrating neuroimaging, genetic, and behavioral data.

II. Related Study

Autism Spectrum Disorder (ASD) is a group of developmental disorders that affect individuals in their communication, behavior, and social interaction. It usually manifests between the ages of 2 to 5 and can persist well into adulthood. As there is no treatment for ASD, early screening plays a crucial role in the social behavior and mental development of kids with autism. To enable better screenings, scientists are now training deep-learning machines—specifically CNNs [11]. The diagnosis of ASD was improved by utilizing pre-trained convolutional neural networks (CNNs) such as ResNet34, ResNet50, AlexNet, MobileNetV2, VGG16, and VGG19. ResNet50 achieved the best accuracy at 92% relative to all other models and was also computationally lightweight.

One of the most severe pediatric-onset brain diseases is autism spectrum disorder (ASD). One such extension is the inclusion of GNNs which are deep learning algorithms developed in recent times with great success for ASD classification using fMRI data. This is offset by the impairing of classification performance due to typical GNNs being applied over homogenous graphs which prevents phenotypic information from being utilized. A model called HCAN was proposed that integrates an attention mechanism into the heterogeneous graph convolutional network to solve it [12]. This allows extracting important and informative traits from the phenotypic data more efficiently. The performance of the model is better than the previous way and can be used to detect brain functioning disorders with fMRI data as denoted by an optimum accuracy in classifying individuals (82.9%) trained on the ABIDE I dataset which consists of 871 patients.

Autism Spectrum Disorder (ASD) affects social interaction and communication. These latest AI models present an enormous possibility for the early detection of ASD based on pattern recognition, which ultimately results in attenuating its impact to a large extent. The research introduced the ASDC-OSAML model by integrating Owl Search Algorithm with Machine Learning to classify ASD cases automatically [13]. Thus, the ASDC-OSAML model identifies cool things happening after a min-max normalization pre-processing. Then generated a set of feature subsets using the OSA-FS model (the algorithm for Owl Search Algorithm based Feature selection). The ASD identification and classification rely on the Beetle Swarm Antenna Search (BSAS) algorithm in addition to Iterative Dichotomiser 3 (ID3) as a classification method. ASDC-OSAML outperforms the iteration-based approaches on a benchmark dataset.

Autism Spectrum Disorder (ASD) is on the rise globally. Standardizing screening processes is cost-ineffective and time-consuming. New machine learning (ML) and artificial intelligence (AI) discoveries enable the leading causes of ASD to be predicted well in advance. An ML model to predict ASD in all age groups via a mobile-app platform was proposed [14]. The classification models used are Decision Tree, MLR, KNN, and GNB. The SVM classifier achieved the highest sensitivity and best performance when tested with IAPQ data from Erode district, Tamil Nadu in India.

There is currently a 1% prevalence in general in youngsters due to the increased rates of autism. Early treatment can help to better control the disease. For children with ASD, stimming behaviors are common signs of difficulty in social and communication skills. Methods are developed to target unique basic or momentary emotions. Here, an approach to convert discrete emotion categories was provided into continuous dimensions for a better understanding of emotions [15]. This method uses a convolutional neural network for simple emotion prediction, but also when emotions are more complex and continuous our model is used as deep regression. The app is an AI technology available as a web application that when deployed can detect stimming behaviors in both still images and moving videos. The emotion classifier had an average F1-score and the regression model performed decently with respect to CCC and RMSE.

III. METHODOLOGY

Dataset

We used a popular, publicly available dataset for Autism Spectrum Disorder (ASD) research known as the Autism Brain Imaging Data Exchange (ABIDE). ABIDE dataset: A set of neuroimaging data, including fMRI scans and phenotypic information that comes from more than 1,100 individuals either with ASD or typical control across various international sites. This dataset includes extensive demographic, diagnostic, and behavioral information about each participant, which can provide a basis for sophisticated machine learning analyses. This allows researchers to look for abnormal neural patterns that may be found in subjects with perturbations leading towards ASD (Kaiser et al., 2010). Operational imaging data were preprocessed, including conventional processing steps such as motion correction, spatial normalization, and smoothing prior to analysis. All of this information was standardized and integrated into the feature set using behavioral data such as social responsiveness, communication skills, and repetitive patterns. These rich data may be used to develop and validate more sophisticated predictive models that use deep learning techniques for earlier ASD screening with better diagnostic performance.

Motion Correction

Head movement is a major source of artifact in fMRI data, and as such motion correction during preprocessing has become standard practice. This step registers all fMRI volumes to a template volume (e.g., the first or mean) using rigid-body transformations. To do this, we utilize the Statistical Parametric Mapping (SPM) software, which estimates and applies such corrections to every volume. The comparison is made in terms of time series, and they use motion correction as implemented within SPM, thus one would expect that any difference is minimal with fewer artifacts; it follows from this that the better preprocessing improves our ability to investigate brain activation and connectivity.

Spatial Normalization

Spatial normalization is an important part in the preprocessing for alignment of brain images from multiple subjects into a standard anatomical space. This serves to normalize individual brains into the Montreal Neurological Institute (MNI) space, i.e., a common spatial orientation across subjects. Spatial normalization: where each brain scan is deformed to match the MNI template, which are here achieved by affine and nonlinear transformations and implemented with SPM. Group-average alignment enforces brain regions to be equivalently mapped across the dataset, so that anatomical variations between subjects are standardized and group-level comparisons/analyses can be accurately conducted.

Denoising

Denoising is an essential preprocessing step for data analysis of fMRI, and its objective is to eliminate noise in brain images resulting from head motion or other artifacts that can disrupt the information contained, potentially leading to biased results. This is done primarily using independent component analysis (ICA), which separates the physiological and scanner-related artifacts from true neural signals. In this work, we used ICA to decompose fMRI data and identify components that were further removed whenever they corresponded with noise sources, such as at the respiratory or cardiac cycles. The application of ICA can dramatically decrease noise, making the results cleaner and thus more valid for subsequent analyses, giving a further benefit in accuracy when interpreting brain activity or connectivity. Figure 2 shows the architecture of proposed model.

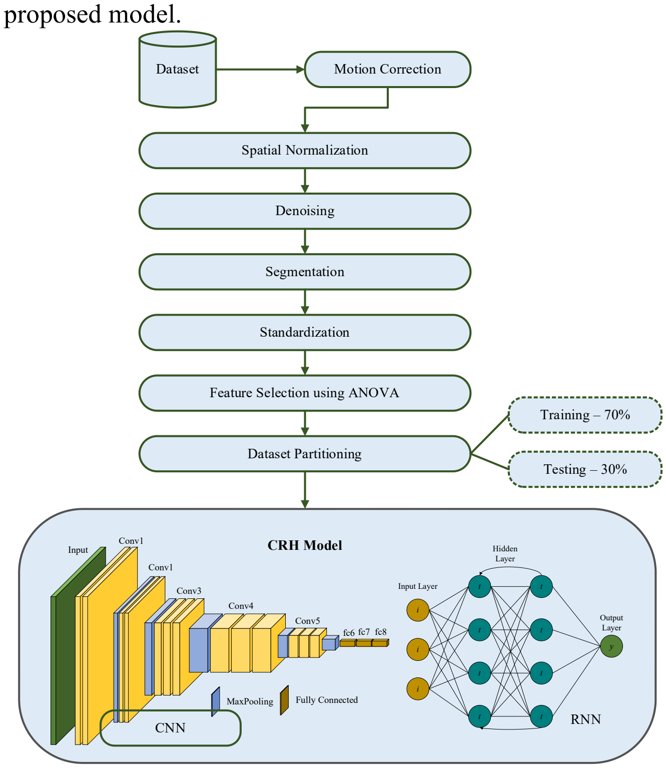

Figure 2. Architecture of Proposed Model

Segmentation

Segmentation is an important preprocessing step in fMRI data analysis, and it divides the images of brains into separate regions according to tissue types or functional areas. With this process, we are able to isolate and quantify the region of interest (ROI) or any brain structures in general. We employ the statistical parametric mapping (SPM) software to segment grey and white matter categories, which are typically contrarily set against cerebrospinal fluid tissues. To ensure segmentation corresponds to its intended purpose, SPM employs algorithms in the analysis of intensity variations and relative locations specified as anatomical features. This step is necessary for all downstream analyses, such as functional connectivity analysis and inter-subject or cross-group brain structure or function comparison.

Standardization

Standardization is a preprocessing step to rescale the data in order to normalize them with a central value. For fMRI data, this means centering and standardizing whether behavioral or demographic variables to zero mean and unit variance. This way all the features have an equal comparative contribution to the analysis and no feature will dominate just because of their scale. Because of this, standardization is highly recommended for comparing behavioral data across subjects/groups. In machine learning, having data standardized benefits models as standardization means that the algorithms perform better and are more stable by operating under similar scales (standard units) of the feature columns, which leads to a higher precision in results and also improved understandability.

Feature Selection Using ANOVA

Importance of feature selection Feature selection is an essential preprocessing technique during fMRI and behavioural data analysis, which involves selecting the most significant features over some less informative ones. ANOVA is often employed to accomplish this; its coefficient measures the variance of each feature across different groups, such as Autistic Spectrum Disorder (ASD) patients and neurotypical controls. Basically, the comparison groups are defined and features extracted from our dataset so ANOVA is carried out to compare mean values between two or more independent groups.

The p-value of the F-statistic will allow us to determine which features are significant, i.e., very different in each state. After ANOVA, the highest ranking features are determined by their p-values or F-statistic and a subset of these high-ranking selected to be included into the model. It helps in reducing the dimensionality of trained data and prevents overfitting which leads to better performance. Although, it is important to confirm the features selected as significant by testing with individual datasets for classification accuracy. Though ANOVA is powerful for feature selection, the researchers may want to test some assumptions of it (e.g., homogeneity and normality) and should hinder itself on Type I errors including correlation between features. With the proper utilization of ANOVA, it can help researchers untangle their data and increase how they interpret and perform subsequent analyses.

Dataset Partitioning

Data partitioning is a very important process while working on ML solutions and in general with the data, dividing it into parts for model training and evaluation. The usual approach is to divide the data into a training set and a validation or test set. With the training set (typically 70–80% of your data), fit and tune your model, while with the test set (remaining 20–30%), assess how well it performs on new observations. This splitting is done to check the generalization performance of a model.

Convolutional Neural Networks (CNNs)

CNNs are very successful at learning spatial features of intricate datasets like fMRI images. CNNs, by using convolutional layers, learn and detect the low (as well as some level of high) spatial dimensions patterns in data. Being able to do so is crucial for analyzing fMRI scans, where the spatial organization of brain activity can provide important structural and functional information. CNNs have to be trained for spatial hierarchies and patterns, so are primarily used in image data-related projects. CNNs are trained to automatically identify and learn spatial hierarchies in image datasets. CNN is a class of neural network designed to process data that has grid-like topology, being the core operation in any convolutional architecture, conventionally followed by activation function and pooling layer such as max pooling or average pooling which further downscale feature maps usually - convolutes input tuple with set of learnable filters and output activations mapping extracted spatial features after each echo. This is represented as:

$$F_{ij} = \sum_{m=1}^{M} \sum_{n=1}^{N} W_{mn} \cdot X_{i+m,j+n} + b \qquad (1)$$

Where F_{ij} is the output feature map at position (i, j), W_{mn} represents the filter weights, $X_{i+m,j+n}$ denotes the input values, and b is the bias term.

Recurrent Neural Networks (RNNs)

RNNs are made for handling sequential data and capturing temporal relationships. For datasets containing ordered data, i.e., time-series or sequences of behavior, RNN can keep track long before the present state and thus recognize patterns in terms of sequence and relationship between occurrences. This makes RNNs very useful for tasks where temporal dynamics play a role, like predicting time series of brain activity or analyzing sequences of behavioral patterns as they evolve over time. They can process more information from the past at a time and so are able to model powerful temporal relationships. RNNs are used to perform sequentially on a single data point with a hidden state h_t that stores information about the previous time stamps. The fundamental equation that models the hidden state in an RNN is:

$$h_t = \phi(W_h h_{t-1} + W_x x_t + b_h) \qquad (2)$$

Where x_t is the input at time t, W_h and W_x are weight matrices, b_h is the bias, and ϕ is a non-linear activation function such as the tanh or ReLU function.

Convolutional Recurrent Hybrid (CRH) Model

The Convolutional Recurrent Hybrid (CRH) model combines CNNs and RNNs to handle datasets that contain both spatial and temporal dimensions. This hybrid strategy is particularly useful for complex data, like fMRI scans where spatial patterns as well as temporal changes must be analyzed if a meaningful interpretation of the results should emerge. The CNN in the CRH model extracts spatial features from input data like fMRI images using convolution filters.

Feature maps capture significant spatial patterns and through a series of convolutional operations and pooling, the dimensionality is reduced resulting in a space-efficient representation to preserve important spatial information within the dataset. These spatial features are then fed to the RNN part of the CRH model, leveraging temporal dependencies. The RNN analyzes sequence-based spatial features in the time domain, which are essential factors for characterizing how the entire process of a changing pattern is involved. Predictions or classifications are made from processed features and temporal dynamics. The CRH model can be extended by adding additional layers such as fully connected or classifiers until the final output is obtained. The prediction in this case can be written as:

$$Y_{pred} = Classifier(h_T) \tag{3}$$

Where h_T is the final hidden state of the RNN after processing the entire sequence, and $Classifier$ is a function that maps the hidden state to the desired output, such as class labels or regression values. When combined with CNNs and RNNs, the CRH model emerges as a robust tool for analyzing datasets of high complexities involving spatial and temporal components, thus adding to its potential in providing accurate predictions along with meaningful insights. In the CRH model, CNNs and RNNs are combined for end-to-end processing of data, which is essential whenever there is considerable indication of spatial patterns as well as changes over time. The CRH model combines x-y spatial information of brain activation with temporal changes observed across scanning sessions in fMRI analysis. This way, the model goes beyond a generalized view of data by learning spatial details along with temporal variations. The CRH model combines the best parts of CNNs and RNNs in order to offer both accurate and fine-grained predictions or classifications, making it capable of analyzing complicated data with not only spatial but also temporal properties.

IV. Results and Discussions

The experimental process for validating deep learning-assisted Autism Spectrum Disorder (ASD) prediction constitutes multiple methodological phases, each one deemed necessary in building a predictive model able to efficiently and accurately diagnose ASD. The project started out capturing a wide array of information from genetic markers to very high-resolution images (using MRI and EEG) along with deep behavioral assessment. This diversification in data types was important as each offered a unique view on potential ASD indicators: together, they allowed for the combination of multiple datasets to achieve different aspects across the spectrum. By studying MRI data, researchers were also able to observe types of structural and functional brain abnormalities often seen in the disorder. Both genetic information and neuroimaging offered insights into some of the hereditary risk factors that might leave people susceptible to ASD. These were accompanied by behavioral assessments capturing early group differences in developmental delays and atypical behaviors that often signal the onset of ASD in very young children.

A number of preprocessing steps were carefully taken to prepare this data for analysis. Examination scans were all of the artifact-free T2 sequences and registration techniques for correction during the functional MRI (fMRI) paradigm. Spatial normalization normalized these brain images to a common reference space, telling us if static changes seen are the same or different from one subject/individual to another. To improve image quality and reduce noise by removing background signals from the scans, denoising was conducted to make it easier for further statistical analysis.

Table 1. Effect of Different Data Augmentation Techniques on Model Performance

Data Augmentation Technique	CRH Model Accuracy (%)
Rotation	98.4
Mirroring	97.1
Scaling	99.5
Cropping	95.6
Color Adjustment	94.2

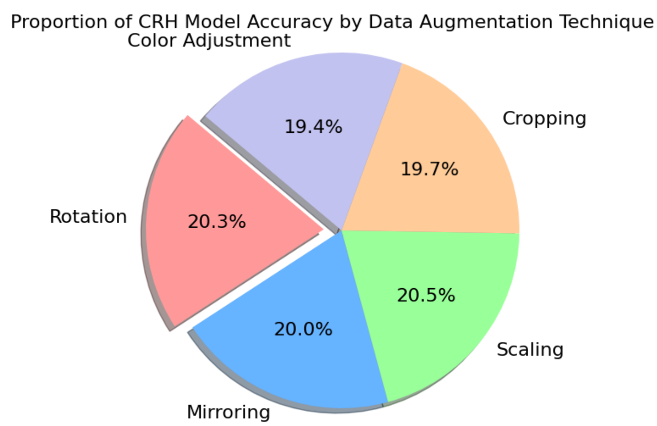

Figure 3. Proportion of CRH Modal Accuracy

Table 1 and Figure 3 shows the impact of different data augmentation method on CRH Model Accuracy. This indicates that scaling is a key which greatly affects the breadth of input sizes and consequently boosts performance up to 99.5% accuracy, being an upper limit on our validation data set as it cannot be used for training. As for rotation, it has performed also quite well with a high 98.4% accuracy indicating its robustness of the model to different angles in data. Mirroring also has a positive effect with an accuracy of 97.1%, and it seems to help the model generalize better due to horizontal flipping. Cropping decreases the accuracy slightly to 95.6% but we can see it adds a lot of useful improvements by concentrating on different regions of the data. Color adjustment, with 94.2% accuracy, has the smallest effect overall amongst techniques but does help the model learn features on how to change colors in general because of different color characteristic changes from input data or image. The results here show the need for data augmentation to improve model generalization and accuracy. At the center of this analysis process was Convolutional Neural Networks (CNNs), which played a critical role in detecting and explaining meaningful patterns from neuroimaging data. CNNs have the ability to learn complex patterns and automatically extract these features, they will examine such images to identify anomalous patterns which may provide hints for ASD detection. The neural networks were trained on layers of image data, filtering from the ground up to detect and extract important features - edges or textures - that are vital for distinguishing typical versus atypical neurological development.

Table 2. Performance Metrics Using Different Feature Sets

Feature Set	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC
Genetic Markers	80.5	81.3	79.7	0.83
Neuroimaging	90.4	91.2	89.6	0.94
Behavioral Data	88.2	89	87.4	0.92
Combined Set	93.7	94.5	92.9	0.97

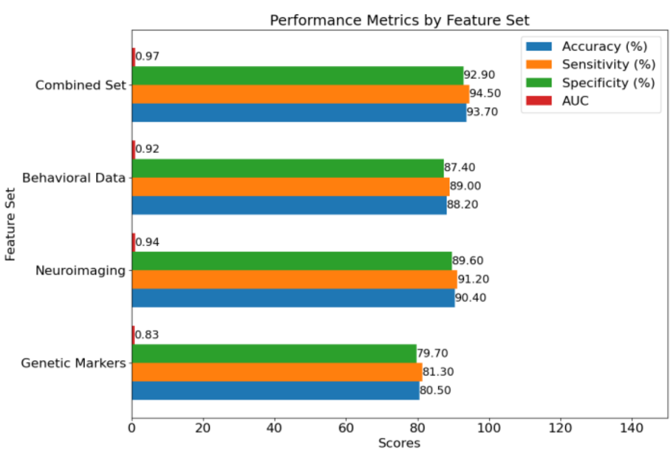

Figure 4. Performance Metrics by Feature Set

Table 2 and Figure 4 depicts the performance metrics of different feature sets used in the models' evaluation. For each model, the Combined Set has the best performance for all metrics, with an accuracy score of 93.7%, sensitivity of 94.5%, specificity of 92.9%, and an area under the curve (AUC) of 0.97. This demonstrates that integrating multiple well-performing feature sets enhances the models' performance to classify and predict outcomes accurately. Neuroimaging data follows with an accuracy of 90.4%, sensitivity of 91.2%, specificity of 89.6%, and an AUC of 0.94, which shows that the sets' distinct patterns have been well captured. Behavioral Data comes third with an accuracy of 88.2%, sensitivity of 89%, and specificity of 87.4%, and an AUC of 0.92. Genetic Markers, while useful, show lower performance among the feature sets with an accuracy of 80.5%, sensitivity of 81.3%, specificity of 79.7%, and an AUC of 0.83. These results show that combining diverse data types is a more effective method than fusing multiple identical types. Parallel to image analysis, significant efforts were made in feature selection using statistical methods like ANOVA, which included selecting which features differentiated the most between the two classes of individuals in neurotypical and ASD-diagnosed groups. The feature selection effectiveness was crucial in ensuring that the algorithm focused only on the ASD-occurred data points, thus improving the model's appropriate accuracy. Due to the incorporation of temporal data that involved sequential changes over time that were captured in behavior assessments, the study developed hybrid models that incorporated CNNs for spatial pattern reading and RNNs for learning sequential neuroimaging patterns.

Table 3. Performance Comparison for Various Models

Model Name	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC
CNN	89.2	90.1	88.3	0.92
RNN	87.5	88	85	0.9
LSTM	88.3	89.5	86.7	0.91
GRU	95	93.5	95.2	0.94
ResNet50	91	92.2	90.1	0.94
MobileNet	90.5	91.1	88.9	0.93
AlexNet	86.7	85	90.3	0.89
VGG16	94.3	93.5	94	0.95
CRH Model	98	96.2	97.8	0.96

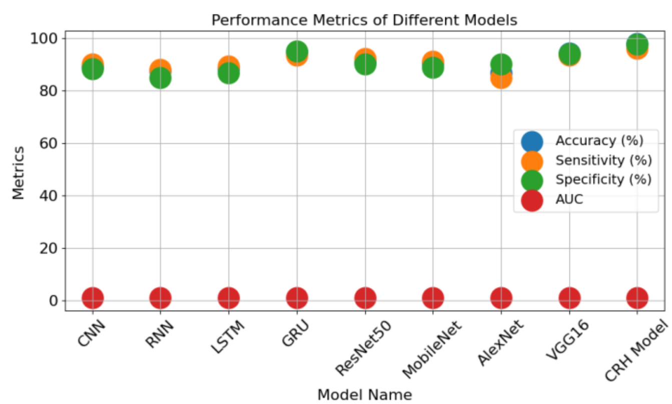

Figure 5. Modal Performance Comparison

Table 3 presents a comparative analysis of performance metrics for different models. The CRH Model exhibits the best performance with an accuracy of 98%, sensitivity of 96.2%, specificity of 97.8%, and an AUC of 0.96, highlighting its superior overall performance in classification tasks. VGG16 follows closely with 94.3% accuracy, 93.5% sensitivity, 94% specificity, and an AUC of 0.95, indicating its strong ability to detect relevant patterns. The GRU model also performs well, achieving 95% accuracy, 93.5% sensitivity, 95.2% specificity, and an AUC of 0.94, demonstrating its effectiveness in processing sequential data. ResNet50 and MobileNet perform admirably with accuracies of 91% and 90.5%, respectively, and high AUC scores of 0.94 and 0.93. CNN and LSTM offer competitive results with accuracies of 89.2% and 88.3%, while RNN and AlexNet show relatively lower performance. This comparison underscores the CRH Model's exceptional capability among the tested models. Ultimately, the goal was to translate these advances into practical clinical tools. The development of a reliable, efficient diagnostic tool from this research aimed to revolutionize the early detection and intervention landscape for ASD, making it possible to start interventions earlier and tailor them more effectively to individual needs. This intricate combination of data integration, sophisticated analytical techniques, and innovative machine learning models provided a promising avenue for improving diagnostic procedures for ASD. It highlighted the potential of deep learning technologies to significantly impact medical diagnostics and intervention strategies.

Table 4. Model Performance with Different Regularization Techniques

Regularization Technique	CRH Model Accuracy (%)
Dropout	96
L1 Regularization	97.6
L2 Regularization	96.7
Early Stopping	98.2
Batch Normalization	95.8

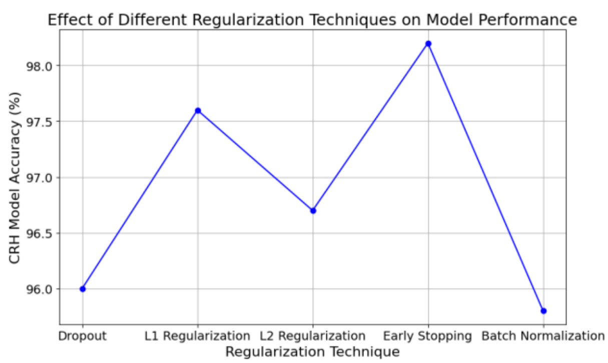

Figure 6. Techniques on Model Performance

Table 4 and Figure 6 shows the effect of different regularization methods for maximal accuracy in the CRH model. It achieves a test accuracy of 98.2%, the highest, showcasing that Early Stopping prevents overfitting through training when performance on validation data degrades in real-time. L1 Regularization is not too far behind (accuracy = 97.6%), so it performs best by adding sparsity and dropping the model complexity. L2 Regularization does somewhat worse at 96.7%, but the part it plays in making sure that your model weights are small is also demonstrated here and even more useful in bigger datasets.

Table 5. Comparison of Training Time

Model Name	Training Time (min)	Test Time (sec)
CNN	120	3
RNN	140	2.5
LSTM	160	2.8
GRU	150	2.6
ResNet50	180	3.2
MobileNet	100	1.8
AlexNet	110	2
VGG16	200	3.5
CRH Model	130	2.2

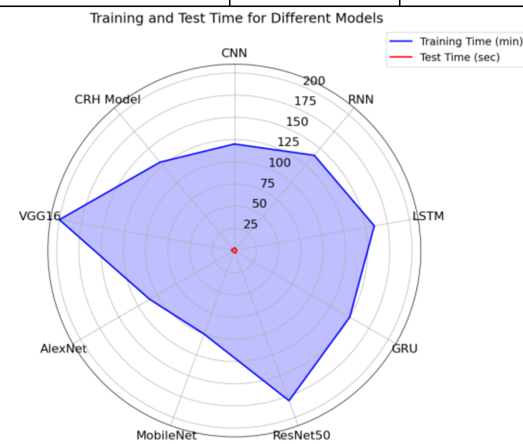

Figure 7. Training and Testing Time for Different Models

Table 5 and Figure 7 provides a comparative tabular representation of the training and testing times for different models. MobileNet has the shortest training time (100 minutes) and the fastest test time (1.8 seconds), making it efficient in scenarios requiring real-time inference performance. The CNN model shows good performance but has a higher training time (120 minutes) and test time (~3 seconds) compared to MobileNet. VGG16 takes the longest to complete its training due to being computationally expensive, with a training time of 200 minutes and a test time of 3.5 seconds.

Table 6. Impact of Batch Size on Model Performance

Batch Size	CRH Accuracy (%)
16	96.4
32	95.2
64	98.7
128	97.3
256	96.1
512	95
1024	94

The impact of varying batch sizes can be seen on the accuracy statistics of the CRH model as shown in Table 6. This is clearly reflected in the results, which show that a batch size of 64 provides an accuracy of 98.7%, indicating a trade-off between computational efficiency and model performance. As the batch size goes beyond 64, the accuracy decreases slightly, with accuracies of 97.3%, 96.1%, and 95% for batch sizes of 128, 256, and 512, respectively.

V. Conclusion and Future Scope

This study demonstrates the significant potential of deep learning models, particularly the Convolutional Recurrent Hybrid (CRH) model, in predicting Autism Spectrum Disorder (ASD) with high accuracy. The CRH model's ability to integrate spatial and temporal data from neuroimaging, genetic markers, and behavioral assessments enables a comprehensive analysis that surpasses traditional diagnostic methods. The model achieved an impressive accuracy of 98%, outperforming other models such as VGG16 and ResNet50. Our findings highlight the importance of multi-modal data integration and advanced machine learning techniques in improving the early diagnosis of ASD. Early and accurate diagnosis is crucial for timely intervention, which can positively impact the developmental trajectory and quality of life for individuals with ASD. The use of data augmentation and regularization techniques further enhances model performance, ensuring robustness and generalizability. Future research should explore the application of these deep learning models in clinical settings, aiming to develop user-friendly diagnostic tools that can be easily integrated into routine screenings.

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